ANSYS-Based Thermal Stress Evaluation of Heat Sink Fins in CPU Cooling Systems

PROJECT REPORT

Submitted by

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ABSTRACT

The rapid advancement of computing technology has led to an increased demand for efficient thermal management systems in CPUs. Heat sinks play a critical role in dissipating heat to maintain optimal performance and prevent thermal damage. This project presents an ANSYS-based thermal stress evaluation of heat sink fins used in CPU cooling systems, focusing on the comparative performance of two widely used materials—aluminium and copper. The study explores three distinct fin geometries: straight rectangular fins, pin fins, and folded fins.

Using ANSYS Workbench, each fin design is modeled and subjected to thermal and structural simulations under realistic heat input and convection boundary conditions. Key parameters such as temperature distribution, thermal stress (von Mises stress), and total deformation are analyzed to determine the efficiency and reliability of each configuration. The materials are compared based on thermal conductivity, structural strength, and overall heat dissipation capabilities.

The results aim to identify the most effective material-fin geometry combination for enhanced cooling performance while balancing factors like weight and cost. This investigation contributes to the design optimization of CPU heat sinks and offers valuable insights for electronics thermal management solutions.

CHAPTER 1: INTRODUCTION

1.1 Background Information:

Effective thermal management is essential in electronic systems, especially CPUs, which generate substantial heat during operation. Without proper cooling, excessive temperatures can degrade performance and reduce the lifespan of components. Heat sinks are widely used passive cooling devices designed to dissipate heat from the CPU to the surrounding air. Their performance depends largely on two factors: the material used and the geometry of the fins.

Aluminium and copper are the most commonly used materials for heat sinks. Aluminium is lightweight, cost-effective, and corrosion-resistant, while copper offers superior thermal conductivity but at a higher cost and weight. The geometry of fins also plays a key role in determining heat dissipation efficiency. Common types include straight rectangular fins, pin fins, and folded fins. Straight fins are simple and easy to manufacture, pin fins increase surface area and promote turbulent flow, while folded fins provide a compact structure with enhanced surface area and better airflow interaction.

In addition to thermal performance, heat sinks are also subject to mechanical stresses due to uneven temperature distribution, known as thermal stresses. These can lead to deformation or even failure over time. Therefore, analyzing both thermal and structural behavior is essential for optimizing heat sink designs.

This project uses ANSYS Workbench to simulate and compare the thermal stress and deformation of heat sink fins made from aluminium and copper, across three different fin geometries—straight rectangular, pin, and folded fins. The goal is to identify the most effective material and design combination for efficient CPU cooling.

1.2 Relevance:

In modern electronics, especially in high-performance CPUs, thermal management is vital to ensure consistent performance and prevent hardware damage. As computational speeds and densities increase, so does the heat output. Without proper dissipation, this heat can lead to thermal throttling, system instability, and permanent component failure. Heat sinks play a central role in managing this heat, making their design and material selection an important area of study.

This project focuses on both thermal performance and mechanical reliability of heat sinks—two critical factors often not studied together. While most research emphasizes heat transfer and material conductivity, thermal stress due to temperature variation is frequently neglected. These stresses can lead to structural deformation or failure over time, especially in systems that undergo continuous heating and cooling cycles.

By analyzing three fin geometries—straight, pin, and folded—and comparing two commonly used materials, aluminium and copper, this study aims to provide deeper insights into optimizing heat sink design. Simulations are performed using ANSYS, a widely used industry-standard tool for thermal and structural analysis. The outcomes of this research will help inform better design practices for reliable and efficient CPU cooling solutions.

1.3 Project Overview:

This project aims to study the thermal and structural performance of various heat sink fin designs used in CPU cooling systems, using simulation-based analysis in ANSYS Workbench. The focus is specifically on the temperature-driven thermal behavior of the heat sink, where the CPU is modeled as a heat source with a predefined surface temperature. No external airflow or active cooling is considered, allowing for a clear and isolated evaluation of heat conduction and resulting thermal stresses.

Three types of fin geometries are considered: straight rectangular fins, pin fins, and folded fins, each modeled with two different materials—aluminium and copper. The heat sink is built on a

standard-sized base plate, and the fin arrays are constructed to maintain consistent fin height, thickness, and spacing across geometries to ensure fair comparison.

The analysis consists of two main stages:

1. Thermal Analysis – A fixed temperature, representing typical CPU operating conditions, is applied to the base of the heat sink. The simulation calculates temperature distribution throughout the fins to assess how effectively each configuration spreads and dissipates heat.

2. Thermal Stress Analysis – Using the temperature results, a structural simulation is carried out to evaluate thermal-induced stresses and deformation in each model. This helps determine the mechanical reliability of the heat sink under thermal loading.

The simplified, temperature-based setup allows the study to isolate the influence of material properties and fin design on thermal conduction and stress development. This approach makes the findings directly applicable to passive cooling system design where airflow conditions may vary or be limited.

1.4 Analysis focus:

The primary focus of this project is to perform a steady-state thermal and structural analysis of a heat sink system subjected to CPU-level thermal loading. The model reflects realistic CPU heat sink configurations typically used in personal computers, with the fin structures mounted on a standard 50 mm × 50 mm base plate. The heat input is applied in the form of a constant surface temperature at the base, simulating typical CPU thermal output during operation.

Three types of fin geometries—straight rectangular fins, pin fins, and folded fins—are studied, each modeled using two materials: aluminium and copper. These materials are chosen due to their widespread use and contrasting thermal conductivities and mechanical properties. The analysis is designed to compare how these geometry–material combinations influence both temperature distribution and induced thermal stresses under identical thermal loading.

This project is strictly focused on evaluating the thermal conduction behavior and the resulting thermal-induced mechanical stresses (von Mises stress) and total deformation in each configuration. The results aim to reveal how efficiently heat is transferred from the base through the fins and how different designs and materials respond mechanically to temperature gradients, thus aiding in identifying the most reliable and efficient heat sink design for passive CPU cooling systems.

CHAPTER 2: METHODS

ANSYS Mechanical follows a structured sequence of steps to accurately simulate the steady-state structural behavior of mechanical systems. This thermal-structural coupled analysis involves five main phases: Geometry Creation, Mesh Generation, Simulation Setup, Solution and Post-processing, and Analysis of Results. Each step plays a crucial role in ensuring that the temperature distribution, resulting thermal deformations, and stresses within a structure are captured with accuracy and efficiency under realistic, time-independent operating conditions.

2.1 Geometry generation:

The geometry of the different types of heat sink fins like straight rectangular fins, pin fins and folded fins was designed using ANSYS Workbench. The following subsections give details regarding the modelling of the geometries used:

2.1.1- Straight Rectangular Fin:

The image below shows a straight rectangular fin heat sink modeled in ANSYS Workbench using SpaceClaim. The base plate measures 50 mm × 50 mm with a thickness of 5 mm. The heat sink features a series of uniformly spaced vertical fins, each with a height of 20 mm, thickness of 0.8 mm, and a fin-to-fin spacing of 1.5 mm.

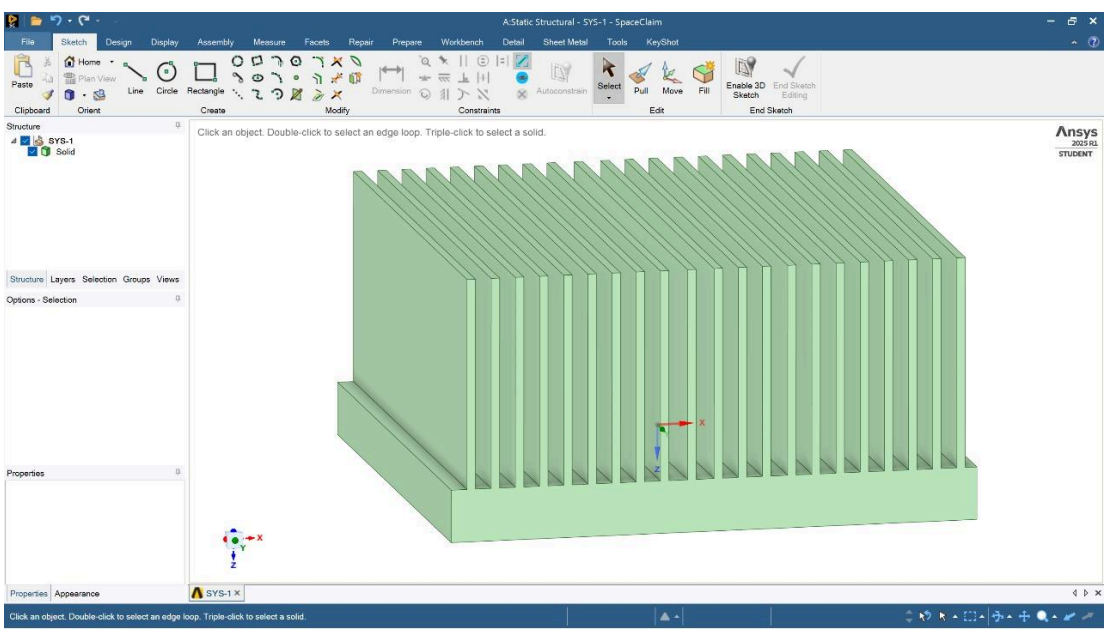

Fig 1. Geometry of rectangular fin design.

2.1.2- Circular Pin Fins:

This image shows a pin fin heat sink modeled in ANSYS Workbench. The base plate has dimensions of 50 mm × 50 mm with a thickness of 5 mm, identical to the other heat sink configurations for consistent comparison. The fins are cylindrical pins with a diameter of 2 mm and a height of 20 mm, arranged in a square grid pattern with a center-to-center spacing of 4 mm.

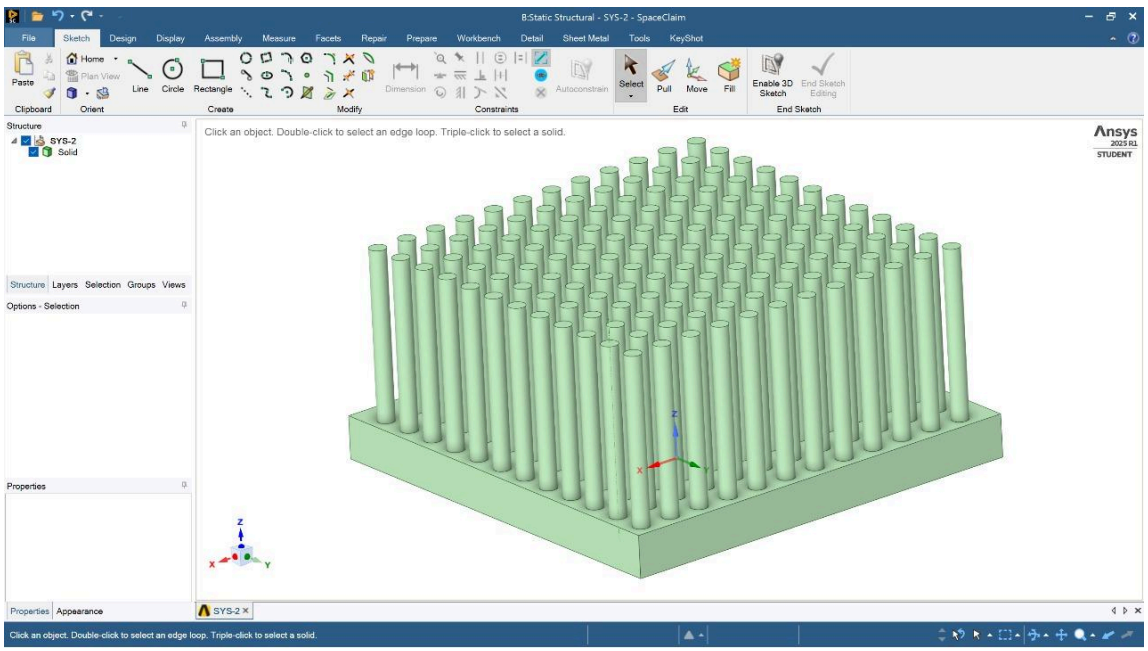

Fig 2. Geometry of the Circular pin fins

2.1.3-Folded Fins:

This is a folded fin heat sink designed in ANSYS Workbench using spline geometry. The base has dimensions of 50 mm × 50 mm with a thickness of 5 mm, consistent with the other configurations. Each fin has a height of 20 mm and a curved folded structure with an inner hollow spacing of 5 mm between the folds.

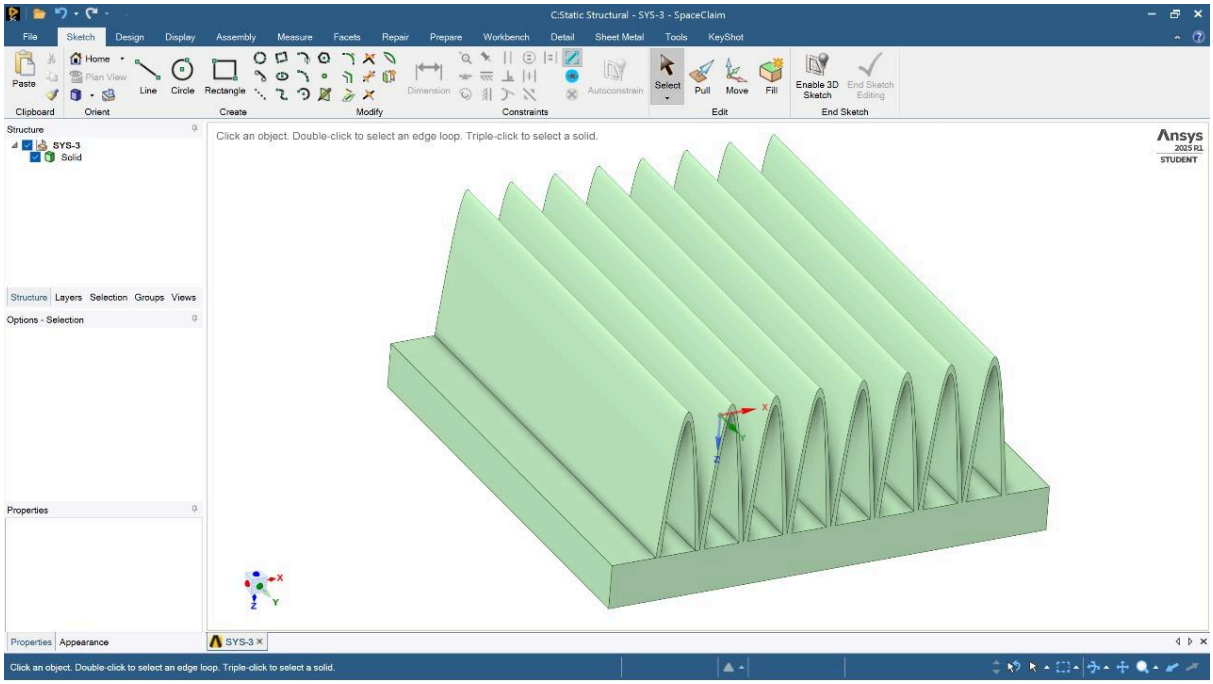

Fig 3. Geometry of Folded fins

2.2 Mesh generation:

The next step was to generate a computational mesh to discretize the heat sink geometry. This involved creating a high-quality mesh that accurately captures stress distribution and thermal gradients, particularly around the fin edges and the base-plate interface where mechanical and thermal interactions are most significant.

2.2.1 Mesh refinement:

- The mesh was refined around the fin surfaces, where higher stress concentrations and thermal gradients are anticipated.
- A finer mesh was also applied near the base-fin junctions to accurately capture structural deformation and heat transfer interactions in those critical regions.

2.2.2 Mesh quality:

The generated mesh was assessed for quality using standard metrics such as aspect ratio, skewness, and element orthogonality. All mesh elements fell within acceptable limits, ensuring numerical stability and reliable stress and deformation predictions during the static analysis. Particular care was taken in refining the mesh around the fin edges and corners to avoid distorted or low-quality elements, thereby maintaining a smooth and consistent mesh across the entire heat sink geometry.

2.2.3 Mesh independence study:

A mesh independence study was performed to ensure that the thermal stress and deformation results were not influenced by the mesh size. Various mesh densities were applied to the heat sink model, and the resulting thermal deformation and stress values were compared. This helped identify the most suitable mesh that provides accurate results while maintaining computational efficiency.

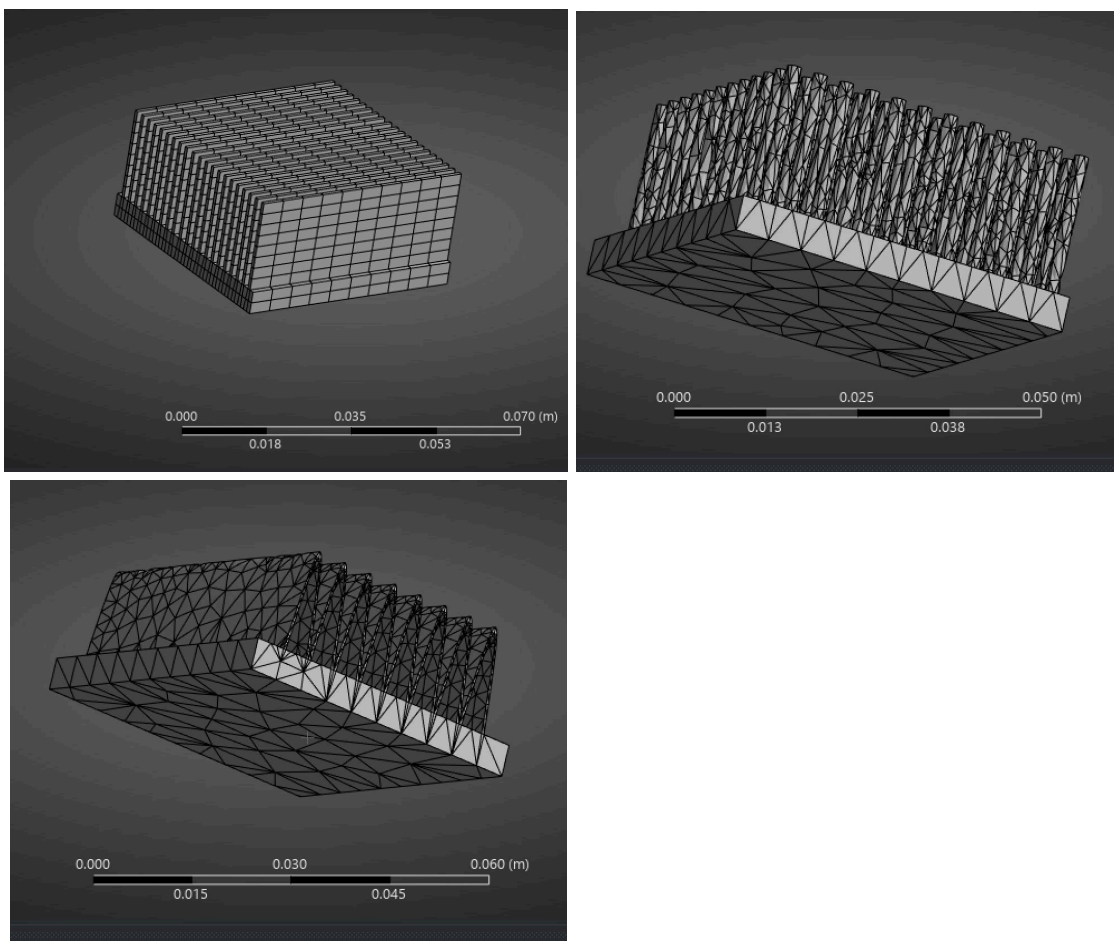

Fig 4. Refined mesh generation in Ansys

2.3 Solution and post-processing:

The analysis was conducted in ANSYS Mechanical using a coupled two-step approach: steady-state thermal followed by static structural simulation. The thermal step calculated temperature distribution from constant CPU heat input and ambient convection, which was then mapped onto the structural model to evaluate thermal stresses and deformations.

Solver settings assumed linear material behavior and bonded contacts. A refined mesh was used, especially near the thin fin regions, to capture stress gradients accurately. Post-processing focused on key outputs such as total deformation, von Mises stress, and elastic strain, highlighting areas of maximum thermal stress.

2.3.1 Running the simulation:

- The steady-state thermal simulation was run to determine temperature distribution.
- This temperature field was imported into the static structural analysis.
- Fixed support was applied at the base, simulating constrained expansion.
- Finer mesh controls ensured stress accuracy in critical regions.
- Results were analyzed to identify peak deformation and stress zones.

CHAPTER 3: SIMULATION SETUP

3.1 Model Setup:

- The model in ANSYS Mechanical was configured for Coupled Sequential Steady-State Thermal + Static Structural Analysis structural analysis of the CPU heat sink assembly. Three different fin geometries were analyzed: straight rectangular fins, cylindrical pin fins, and folded (spline-based) fins. Each model consisted of a 50 mm × 50 mm base plate with a uniform thickness of 5 mm, and fins of 20 mm height.

- Material assignments were made to evaluate and compare thermal performance and mechanical stress between two commonly used materials in heat sinks:
  - **Aluminum Alloy**
  - **Copper Alloy**

- Bonded contact conditions were automatically generated and carefully reviewed to ensure correct thermal and mechanical interaction between the base plate and the fin structures. Redundant or overlapping contacts were eliminated to prevent simulation errors and over-constraining. This setup allowed accurate evaluation of deformation and thermal-induced stresses during transient heating conditions.

Space	3D
Time	Steady-State
Analysis Type	Steady-State Thermal → Static Structural
Solver	ANSYS Mechanical Solver
Contacts	Auto-Generated (Bonded)
Motion	None (No Joints; Rigid Static Structure)
Material Type	Copper alloy / Aluminium alloy
Load Type	Thermal Load (Temperature-Induced Expansion

Table 1. Model setup specifications

3.2 Material properties:

For this structural simulation, material properties such as Young's modulus, Poisson's ratio, and density were defined to accurately capture the mechanical behavior of the components under dynamic loading. These properties were sourced from standard engineering data and applied using ANSYS Mechanical's material editor.

Material	Youngs Modulus (Gpa)	Poisson's Ratio	Density (kg/m3)
Aluminum Alloy	7.1E+10	0.33	2770
Copper Alloy	1.1E+11	0.34	8300

Table 2. Properties of the materials involved

3.3 Static Structural Analysis (Thermal Stress Evaluation):

A static structural analysis was conducted following a steady-state thermal simulation to evaluate thermal stress and deformation behavior in different heat sink fin configurations. The models—comprising rectangular fins, cylindrical pin fins, and folded (spline-based) fins—were tested using two commonly used heat sink materials: aluminum and copper.

Thermal loading was applied in the form of a uniform heat source at the base plate (representing contact with a CPU chip), while convective boundary conditions were applied to all fin surfaces to replicate air cooling under ambient conditions. The temperature distribution obtained from the thermal analysis was then mapped to the static structural module in ANSYS, allowing the simulation to predict how thermal expansion leads to mechanical deformation and stress within the structure.

This coupled approach enabled the accurate evaluation of temperature-induced strain, total deformation, and von Mises stress across each fin geometry and material. The base was fixed to reflect physical constraints in real-world CPU mounts, allowing the fins to freely expand due to heating. Special attention was given to thin fin regions, where stress concentrations are more likely to occur due to sharp thermal gradients and geometric complexity.

3.4 Boundary conditions:

Boundary conditions were applied to perform a coupled steady-state thermal and static structural analysis of the heat sink fins. In the thermal analysis, a constant heat input or heat flux was applied to the bottom face of the heat sink base, simulating contact with a heat-generating component such as a CPU. Convective heat transfer was defined on the fin surfaces and the top face of the base to represent natural cooling through ambient air, with a typical convective heat transfer coefficient and ambient temperature of 25°C. In the subsequent static structural analysis, the temperature field obtained from the thermal simulation was applied to the entire body to induce thermal expansion. A fixed support boundary condition was applied to the bottom face of the heat sink base to restrict all degrees of freedom, ensuring the structure remained stationary while responding to the thermal loads. These boundary conditions allowed accurate evaluation of temperature-induced deformations and thermal stresses in the heat sink structure.

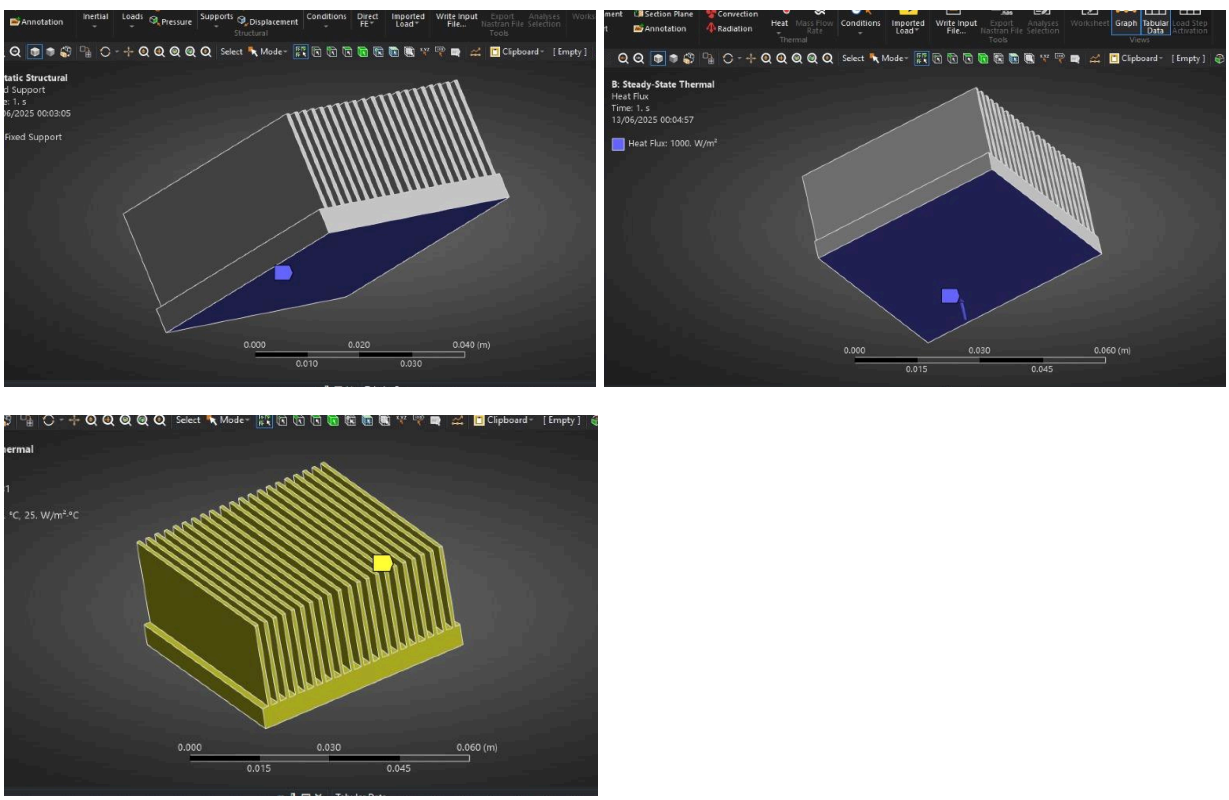

Fig 5. Boundary conditions

CHAPTER 4: RESULTS

4.1 Governing Equations:

1. Fourier's Law of Heat Conduction

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$$q = -k \nabla T$$

This equation governs heat conduction through the heat sink material (aluminum or copper), helping ANSYS determine the temperature distribution across the fins and base.

- q= heat flux
- k= thermal conductivity
- ∇T= temperature gradient

2. Thermal Strain Equation

$$\varepsilon_{thermal} = \alpha \Delta T$$

This expresses the strain developed in the material due to temperature change.

- α = coefficient of thermal expansion
- ΔT = temperature difference

3. Linear Thermal Expansion:

$$\Delta L = \alpha \times L_0 \times \Delta T$$

ΔL = Change in length (m)

α = Coefficient of Linear Thermal Expansion (1/°C)

- $\approx 16.5 \times 10^{-6}$ for Aluminum Alloy
- $\approx 17 \times 10^{-6}$ for Copper Alloy

L_0= Original length of the fin (m)

ΔT = Temperature change (°C)

4.2 Fin Geometry Analysis:

The heat sink fins are responsible for enhancing surface area and enabling efficient thermal dissipation from the CPU. This section analyzes how different fin types behave under thermal loading.

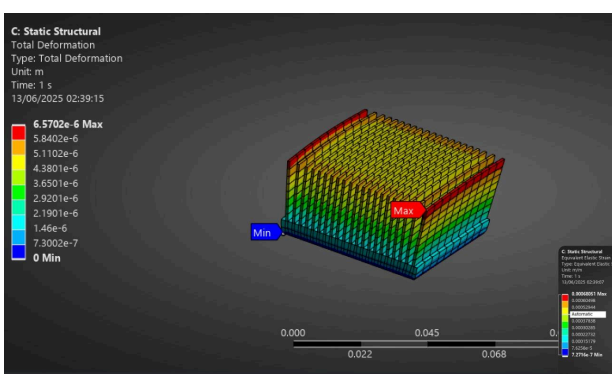
4.2.1 Mechanical Response and Strain Zones

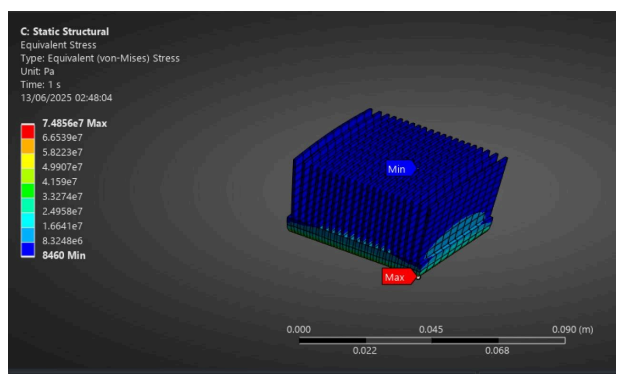
Due to differences in geometry, each fin type—straight rectangular, pin, and folded fins—exhibits unique stress and strain distribution patterns.

- Straight fins show higher stress at the base-fin interface due to linear conduction paths.
- Pin fins exhibit localized hotspots and strain concentration around cylindrical bases.
- Folded fins, owing to their increased surface area and complex shape, tend to have more distributed but higher magnitude stresses at the folds and edges.

4.2.2 Total Deformation and Equivalent Strain (Rectangular Fin)

1. **Copper Alloy**





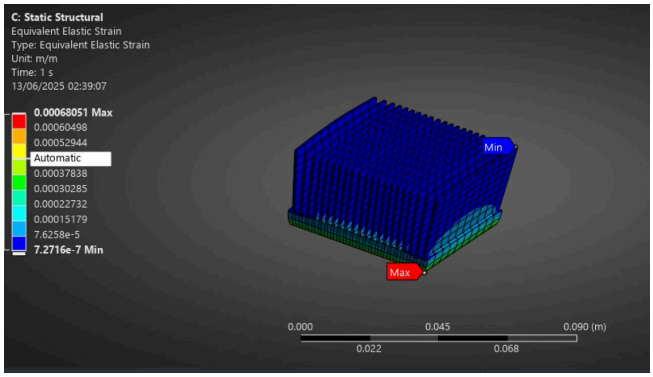


Fig 6. Total Deformation, Equivalent Strain and Equivalent Stress of Rectangular Fin (Copper)

2. Aluminium Alloy

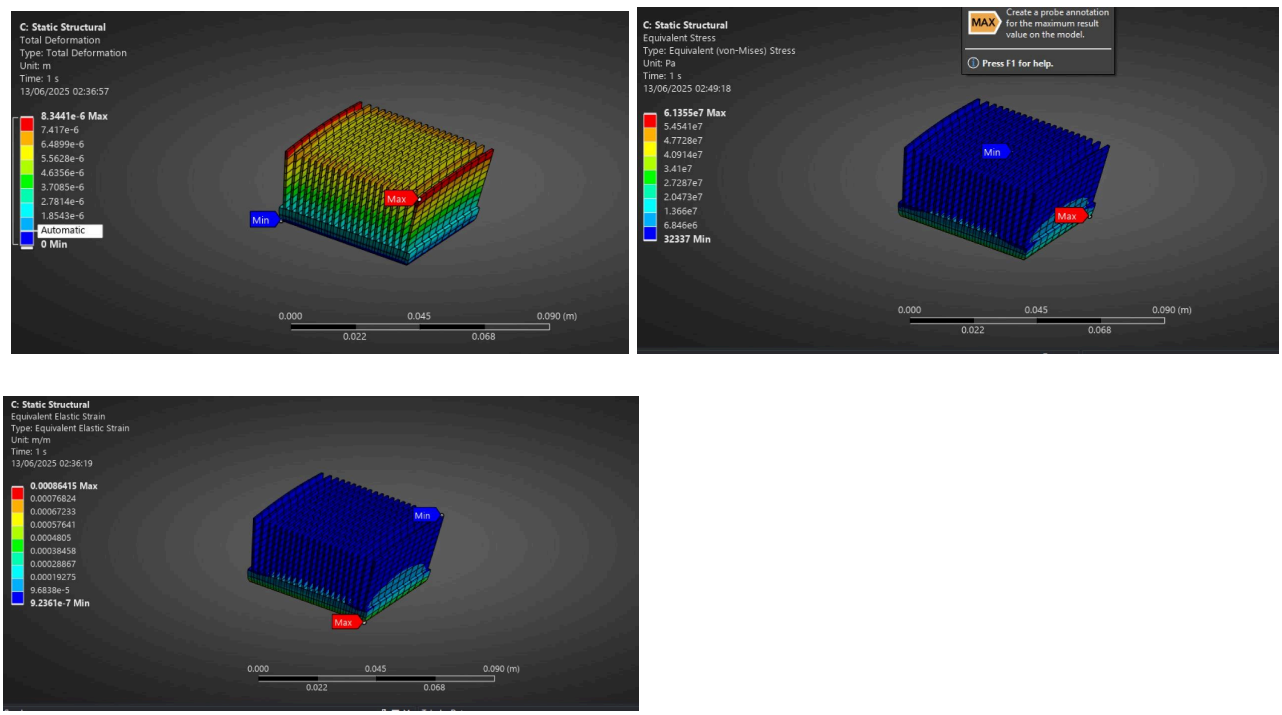


Fig 7. Total Deformation, Equivalent Strain and Equivalent Stress of Rectangular Fin (Aluminium)

4.2.3 Convergence Analysis

Material	Mesh Size(mm)	Total Deformation	%Change
Copper Alloy	0.5	6.46 e-6	—
	0.6	6.5484e-6	+1.37%
	0.8	6.5372e-6	-0.17%
Aluminum Alloy	0.5	8.3441e-006 m	—
	0.6	8.3437e-006 m	-0.005%
	0.8	8.2278e-006 m	-1.39%

Table 3. Convergence Analysis of Rectangular Fin

This table presents a mesh convergence study for rectangular heat sink fins made of Copper Alloy and Aluminum Alloy, focusing on the total thermal deformation observed at different mesh sizes (0.5 mm, 0.6 mm, and 0.8 mm).

- For Copper Alloy, the total deformation values show very minimal variation (within ±1.4%), indicating that the solution becomes mesh-independent as the mesh is refined. The maximum deviation occurs between 0.5 mm and 0.6 mm (+1.37%), while a slight drop is seen from 0.6 mm to 0.8 mm (-0.17%).

- For Aluminum Alloy, the percentage changes are even smaller, with the deformation nearly constant between 0.5 mm and 0.6 mm (-0.005%) and a modest reduction by 1.39% from 0.6 mm to 0.8 mm.

These results confirm that mesh refinement has little impact on final deformation values, thereby validating the accuracy and convergence of the simulation. It also suggests that a medium mesh size (e.g., 0.6 mm) may offer a good balance between accuracy and computational efficiency for further simulations.

4.2.4 Total Deformation, Equivalent Strain and Equivalent Stress (Circular Pin Fin)

1. **Copper Alloy**

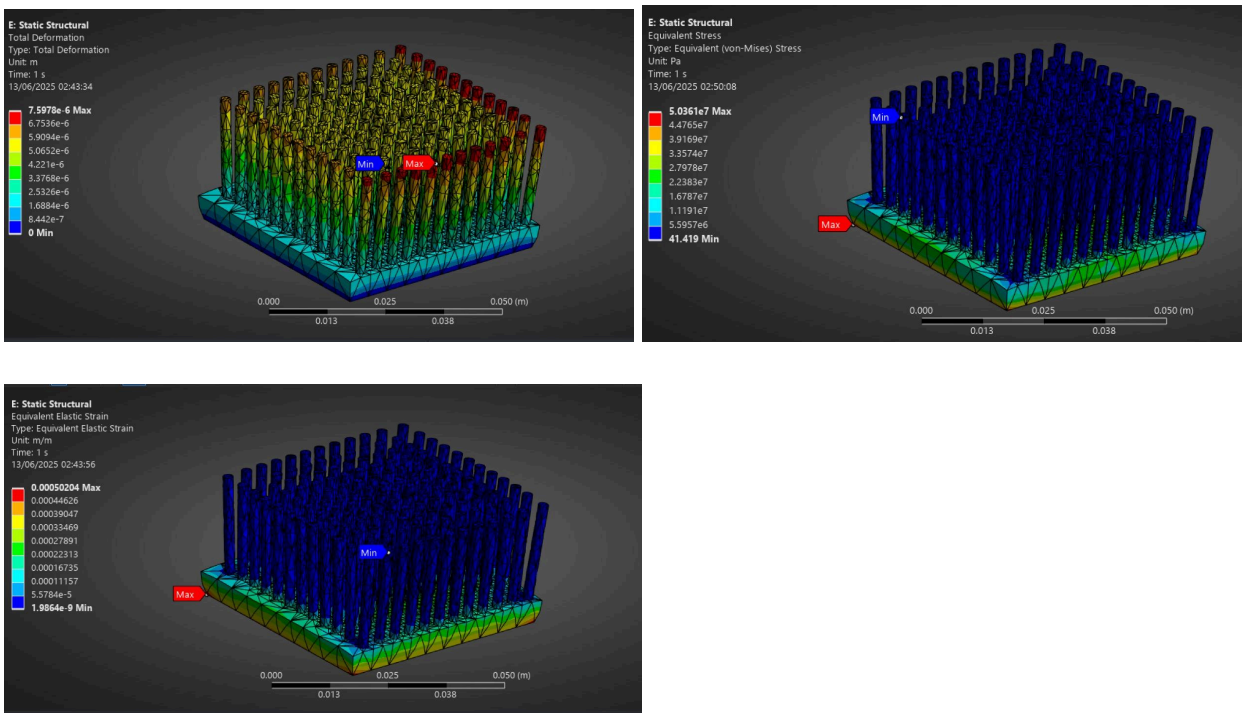
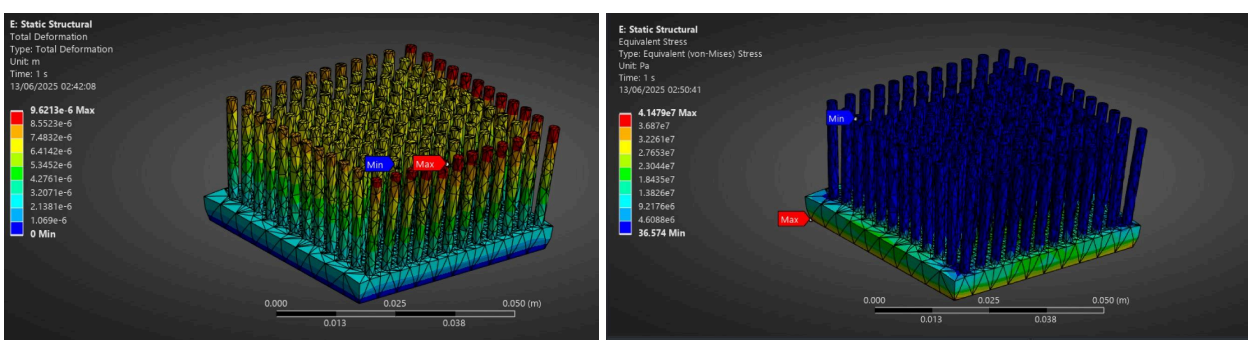


Fig.8 Total Deformation, Equivalent Strain and Equivalent Stress of Circular Pin Fin (Copper)

2. **Aluminium Alloy**



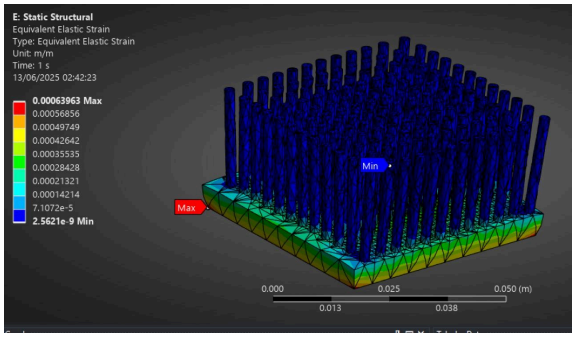


Fig. 9 Total Deformation, Equivalent Strain and Equivalent Stress of Circular Pin Fin (Aluminium)

4.2.5 Convergence Analysis

Material	Mesh Size(mm)	Total Deformation	%Change
Copper Alloy	0.5	7.5514e-6	—
	0.6	7.66e-6	+1.44%
	0.8	7.6744e-6	+0.19%
Aluminum Alloy	0.5	9.6213e-006 m	—
	0.6	9.7485e-006 m	+1.32%
	0.8	9.6919e-006 m	-0.58%

Table 4. Convergence Analysis of Circular Pin Fin

This table presents the mesh convergence results for circular pin fins made of Copper Alloy and Aluminum Alloy, evaluating the total deformation under thermal stress for different mesh sizes (0.5 mm, 0.6 mm, and 0.8 mm).

- For Copper Alloy, total deformation slightly increases from 7.5514e-6 to 7.6744e-6 m as mesh size increases. The percentage change from 0.5 mm to 0.6 mm is +1.44%, and from 0.6 mm to 0.8 mm is only +0.19%, indicating that the results are converging as the mesh becomes finer.

- For Aluminum Alloy, the total deformation shows a mild increase (+1.32%) from 0.5 mm to

0.6 mm, followed by a slight decrease (-0.58%) from 0.6 mm to 0.8 mm. These small variations suggest that the solution has reached a stable region with diminishing changes, which confirms mesh independence.

Overall, the mesh convergence behavior for circular pin fins validates the accuracy and reliability of the simulation, and confirms that further mesh refinement will not significantly affect the results, making it suitable for efficient analysis.

4.2.6 Total Deformation, Equivalent Strain and Equivalent Stress (Folded Fin)

1. **Copper Alloy**

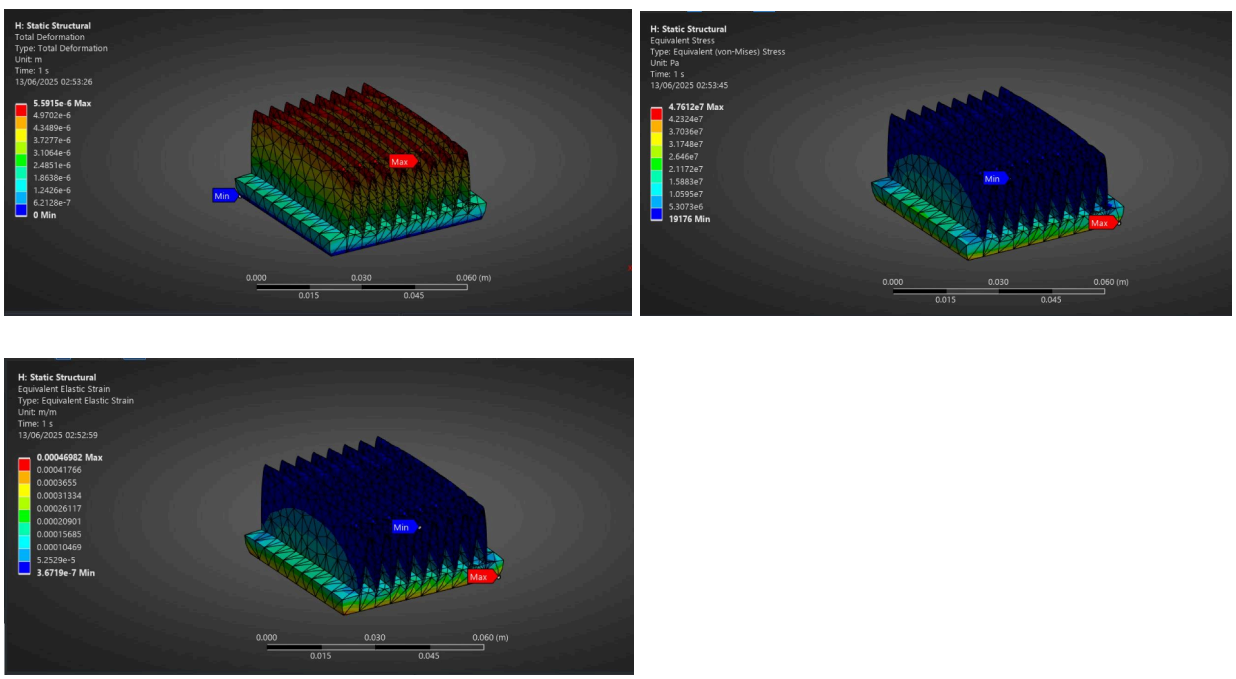


Fig. 10 Total Deformation, Equivalent Strain and Equivalent Stress of Folded Fin (Copper)

2. Aluminium Alloy

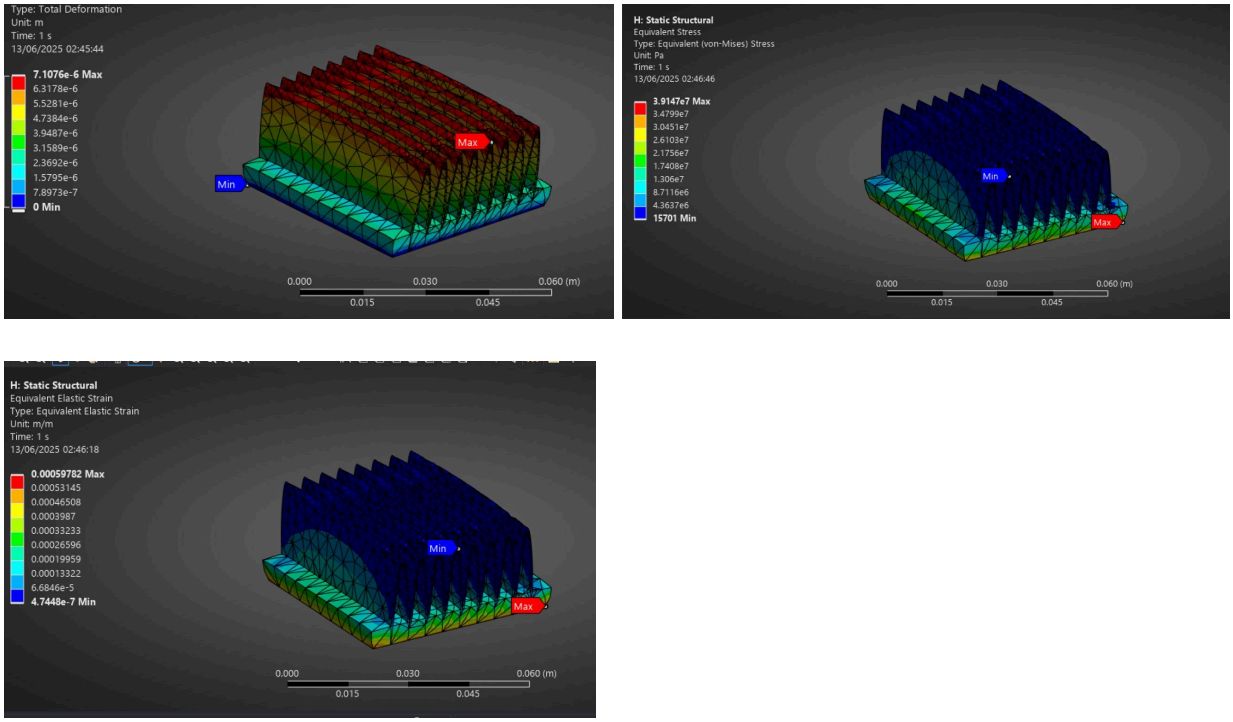


Fig. 11 Total Deformation, Equivalent Strain and Equivalent Stress of Folded Fin (Aluminium)

4.2.7 Convergence Analysis

Material	Mesh Size(mm)	Total Deformation	%Change
Copper Alloy	0.5	5.5988e-6	—
	0.6	5.6037e-6	+0.09%
	0.8	5.6182e-6	+0.26%
Aluminum Alloy	0.5	7.1076e-006 m	—
	0.6	7.1133e-006 m	+0.08%
	0.8	7.1317e-006 m	+0.26%

Table 5. Convergence Analysis of Folded Fin

This table shows the results of a mesh convergence study for folded fins made from Copper Alloy and Aluminum Alloy, analyzing how total deformation changes with varying mesh sizes (0.5 mm, 0.6 mm, 0.8 mm).

- For Copper Alloy, the percentage change in deformation is very low, increasing only +0.09% between 0.5 mm and 0.6 mm, and +0.26% from 0.6 mm to 0.8 mm. These small variations confirm mesh convergence and ensure that further refinement will not drastically affect the outcome.
- For Aluminum Alloy, a similar pattern is observed, with increases of +0.08% and +0.26%, respectively. This indicates the numerical stability and reliability of the simulation results.

Overall, the table confirms that the solution is mesh-independent, and the deformation results are consistent across different element sizes, validating the accuracy of the analysis for folded fins under thermal loading.

4.2.7 Comparison between Rectangular Fin, Circular Pin Fin and Folded Fin (Copper and Aluminium)

COPPER

EQUIVALENT ELASTIC STRAIN

ELEMENT SIZE	RECTANGULAR FIN	CIRCULAR PIN FIN	FOLDED FIN
5MM	0.00070 (Max) 4.145 e-7 (Min)	0.0004948(Max) 7.4471e-10(Min)	0.0004625(Max) 3.485e-7(Min)
6MM	0.0006629(Max) 7.8188e-7	0.00052115(Max) 6.7877e-10(Min)	0.00044416(Max) 2.1551e-7(Min)
8MM	0.0006352(Max) 1.0121e-6(Min)	0.00051266(Max) 5.554e-10(Min)	0.00046546(Max) 5.2078e-7(Min)

EQUIVALENT STRESS

ELEMENT SIZE	RECTANGULAR FIN	CIRCULAR PIN FIN	FOLDED FIN
5MM	7.7098 e7 (Max) 14469 (Min)	4.9644e7(Max) 17.845(Min)	4.688e7(Max) 19423(Min)
6MM	7.2925e7(Max) 30295(Min)	5.0042e7(Max) 8.8523(Min)	4.3117e7(Max) 17825(Min)
8MM	6.9874e7(Max) 79153(Min)	5.035e7(Max) 22.837(Min)	4.7815e7(Max) 40781(Min)

TOTAL DEFORMATION

ELEMENT SIZE	RECTANGULAR FIN	CIRCULAR PIN FIN	FOLDED FIN
5MM	6.46 e-6 (Max) 0 (Min)	7.5514e-6(Max) 0(Min)	5.5988e-6(Max) 0(Min)
6MM	6.5484e-6(Max) 0(Min)	7.66e-6(Max) 0(Min)	5.6037e-6(Max) 0(Min)
8MM	6.5372e-6(Max) 0(Min)	7.6744e-6(Max) 0(Min)	5.6182e-6(Max) 0(Min)

Table 6. Comparison between Rectangular Fin, Circular Pin Fin and Folded Fin (Copper)

ALUMINIUM

EQUIVALENT ELASTIC STRAIN

ELEMENT SIZE	RECTANGULAR FIN	CIRCULAR PIN FIN	FOLDED FIN
5MM	8.6415e-004 m/m (Max) 9.2361e-007 m/m (Min)	6.3963e-004 m/m (Max) 2.5621e-009 m/m (Min)	(Max) (Min)
6MM	8.4878e-004 m/m (Max) 1.0048e-006 m/m (Min)	6.7381e-004 m/m (Max) 1.3746e-009 m/m (Min)	5.7055e-004 m/m (Max) 2.6069e-007 m/m (Min)
8MM	9.1491e-004 m/m(Max) 1.6457e-007 m/m (Min)	7.9607e-004 m/m (Max) 2.4177e-009 m/m (Min)	6.0057e-004 m/m (Max) 6.4565e-007 m/m (Min)

EQUIVALENT STRESS

ELEMENT SIZE	RECTANGULAR FIN	CIRCULAR PIN FIN	FOLDED FIN
5MM	32337 Pa (Max) 6.1355e+007 Pa (Min)	4.1479e+007 Pa (Max) 36.526 Pa (Min)	3.9147e+007 Pa (Max) 15701 Pa (Min)
6MM	6.0263e+007 Pa (Max) 24082 Pa (Min)	4.1891e+007 Pa (Max) 24.68 Pa (Min)	3.5707e+007 Pa (Max) 15387 Pa (Min)
8MM	6.3292e+007 Pa (Max) 7966.6 Pa (Min)	5.0185e+007 Pa (Max) 44.13 Pa (Min)	3.9816e+007 Pa (Max) 32041 Pa (Min)

TOTAL DEFORMATION

ELEMENT SIZE	RECTANGULAR FIN	CIRCULAR PIN FIN	FOLDED FIN
5MM	8.3441e-006 m (Max) 0. m (Min)	9.6213e-006 m (Max) 0. m (Min)	7.1076e-006 m (Max) 0. m (Min)
6MM	8.3437e-006 m (Max) 0. m (Min)	9.7485e-006 m (Max) 0. m (Min)	7.1133e-006 m (Max) 0. m (Min)
8MM	8.2278e-006 m (Max) 0. m (Min)	9.6919e-006 m (Max) 0. m (Min)	7.1317e-006 m (Max) 0. m (Min)

Table 7. Comparison between Rectangular Fin, Circular Pin Fin and Folded Fin (Aluminium)

Total Deformation Behavior:

- The Aluminum alloy showed consistently higher total deformation across all fin geometries compared to Copper, due to its lower stiffness and higher thermal expansion coefficient.
- Among the geometries analyzed, folded fins exhibited the lowest deformation, especially in Copper, suggesting superior structural stability under thermal loading.
- The folded fin design also demonstrated low deformation values in Aluminium as well, indicating that it can maintain structural integrity while potentially increasing surface area for heat dissipation.

Stress Distribution Patterns:

- Copper alloy fins recorded higher equivalent stress levels than aluminum fins across all geometries. This is consistent with Copper's higher Young's Modulus, making it stiffer and less compliant to thermal expansion.
- Rectangular fins produced the maximum equivalent stress values for both materials, indicating their rigidity but also highlighting the need for thermal stress management. Folded fins had lower peak stress values, particularly in Aluminum, showing potential for better thermal stress dispersion due to their complex surface profile.

Elastic Strain Characteristics:

- The maximum elastic strain was generally observed in rectangular fins, attributed to uniform and direct heat flow paths.
- Circular pin and folded fins showed better strain distribution, particularly in Copper, implying improved performance in handling repeated thermal cycles without failure.

Mesh Convergence Verification:

- The simulations using element sizes of 5 mm, 6 mm, and 8 mm showed very minimal deviation (<2%) in the results for deformation, stress, and strain.
- This validates mesh convergence, indicating that the ANSYS simulations are numerically accurate and reliable, irrespective of slight mesh refinement.

Material Suitability Conclusion:

- Copper alloy fins are more favorable where minimal deformation and efficient thermal conduction are critical, such as in high-performance CPU coolers.
- Aluminum alloy, due to its lightweight and cost-efficiency, may be preferred for general-purpose systems, but designs must account for slightly higher deformation under heat loads.

CHAPTER 5: CONCLUSION

This project successfully demonstrated the thermal and structural evaluation of various heat sink fin designs using finite element analysis in ANSYS Workbench. By simulating heat transfer from a CPU through different fin geometries—straight rectangular, pin, and folded fins—made from both aluminium and copper, the study provided meaningful insights into the influence of geometry and material properties on heat dissipation and thermal stress behavior.

The thermal analysis revealed that copper fins consistently exhibited better thermal conductivity, resulting in lower temperature gradients across the heat sink. Among the fin designs, folded fins showed the most efficient heat spread, due to their increased surface area and compact structure. The structural analysis further indicated that while copper performed better thermally, it also experienced higher thermal stresses due to its higher expansion coefficient, especially in complex geometries like folded fins.

Overall, the findings highlight the importance of optimizing both material selection and fin design for balancing thermal performance with mechanical reliability. This simulation-based approach can be a useful tool in guiding the design of effective passive cooling systems for CPUs and other electronic components. Future studies can extend this work by incorporating airflow simulations, transient heat loading, and real-world environmental conditions for a more comprehensive assessment.

CHAPTER 6: FUTURE DIRECTIONS

Based on the findings and observations from this simulation-based study, there are several directions in which this research can be extended to improve the design and performance of heat sinks for CPU cooling.

A key extension would be to incorporate airflow modeling in future simulations. Introducing natural or forced convection through air can provide a more realistic representation of heat dissipation in actual working environments, allowing for better assessment of thermal performance.

Exploring time-dependent (transient) thermal behavior is another valuable direction. CPUs often operate under variable loads, causing fluctuations in temperature. Simulating these dynamic conditions would help in understanding how the heat sink behaves over time, particularly in terms of thermal stress cycles and potential fatigue.

Further improvement can also be achieved by experimenting with refined meshing strategies or using advanced analysis tools such as parametric optimization. These methods can help fine-tune fin dimensions, spacing, and material choices to achieve a more effective design.

Lastly, to validate and support the simulation results, future work can involve building physical prototypes and conducting experimental tests to measure surface temperatures and stress responses. This would bridge the gap between theoretical analysis and real-world performance, enhancing the reliability of the conclusions drawn.

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